

FIN-220 Hands-On Machine Learning
Final Exam
Due on December 15, 2025

1. What are some of the main applications of clustering algorithms?
2. Describe two techniques to select the right number of clusters when using k-means.
3. What is label propagation? Why would you implement it, and how?
4. Load the MNIST dataset (introduced in Chapter 3) and split it into a training set and a test set (take the first 60,000 instances for training, and the remaining 10,000 for testing). Train a random forest classifier on the dataset and time how long it takes, then evaluate the resulting model on the test set. Next, use PCA to reduce the datasets dimensionality, with an explained variance ratio of 95%. Train a new random forest classifier on the reduced dataset and see how long it takes. Was training much faster? Next, evaluate the classifier on the test set. How does it compare to the previous classifier? Try again with an SGDClassifier. How much does PCA help now?
5. The classic Olivetti faces dataset contains 400 grayscale 64×64 -pixel images of faces. Each image is flattened to a 1D vector of size 4,096. Forty different people were photographed (10 times each), and the usual task is to train a model that can predict which person is represented in each picture. Load the dataset using the `sklearn.datasets.fetch_olivetti_faces()` function, then split it into a training set, a validation set, and a test set (note that the dataset is already scaled between 0 and 1). Since the dataset is quite small, you will probably want to use stratified sampling to ensure that there are the same number of images per person in each set. Next, cluster the images using k-means, and ensure that you have a good number of clusters (using one of the techniques discussed in Chapter 9). Visualize the clusters: do you see similar faces in each cluster?